

# Machine Learning Regression

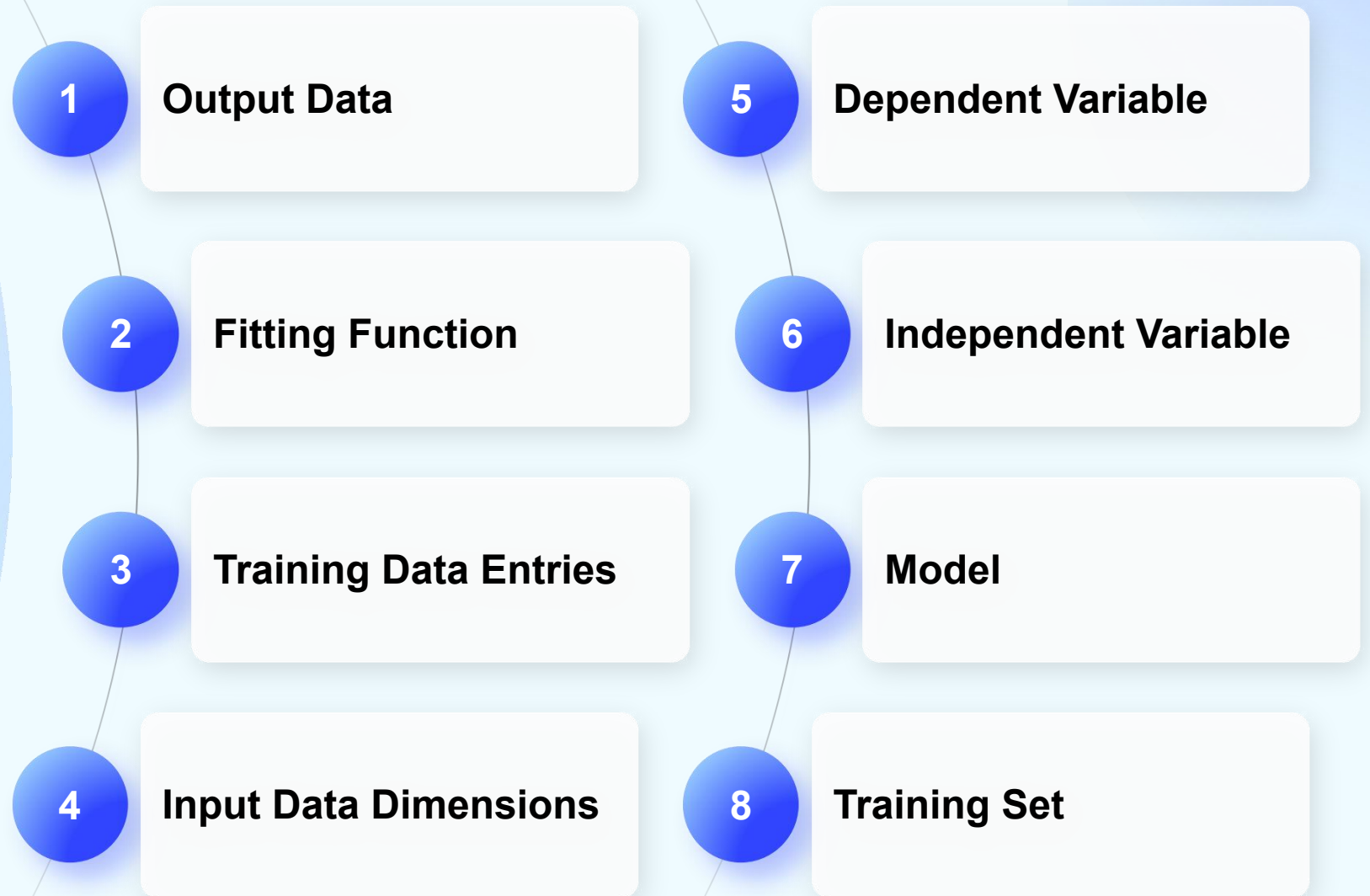


# Definition and Purpose of Regression

- **Predicting Continuous Values:** Regression analysis is a supervised learning method aimed at **predicting continuous numerical outputs** (such as house prices, temperatures, sales, etc.) through input features, rather than discrete classification labels. The core is to establish a mathematical relationship model between features and target variables.
- **Function fitting:** Regression minimizes the error between predicted and true values by fitting the best function (such as linear or polynomial), thereby revealing potential patterns in the data. For example, linear regression assumes that the target variable is the weighted sum of features plus a bias term.
- **Model interpretability:** Compared to complex models such as neural networks, simple regression models such as linear regression have high interpretability and can intuitively display the degree of influence of each feature on the results (such as the positive and negative coefficients and size).
- **Benchmark model:** Regression is often used as a baseline model for machine learning projects to verify whether data contains predictable linear patterns and provide performance comparison benchmarks for subsequent complex models.

## Core Concepts of Regression

# Key Element s of Regress ion



# Linear vs. Non-Linear Regression Landscape

## Goal

Predict a continuous target from features

## Spectrum

Linear  
(simple)



Polynomial  
(non-linear)

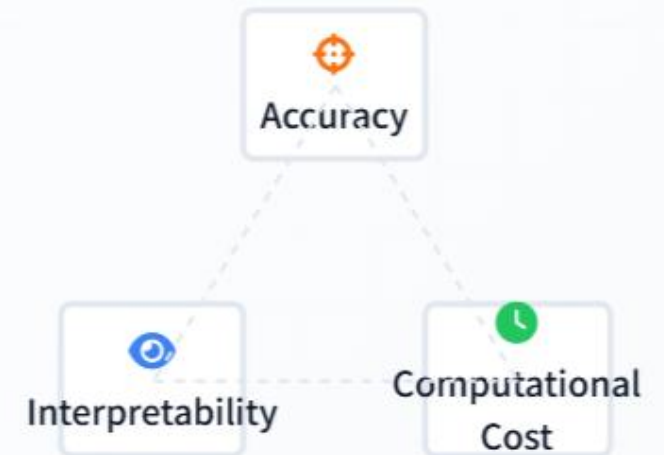


Regularized  
(robust)



Ensemble  
(flexible)

## Trade-off Triangle



# Applications of Regression

## House Price Prediction

Regression can be used to predict house prices based on features like area, number of rooms, location, etc.

## Sales Forecasting

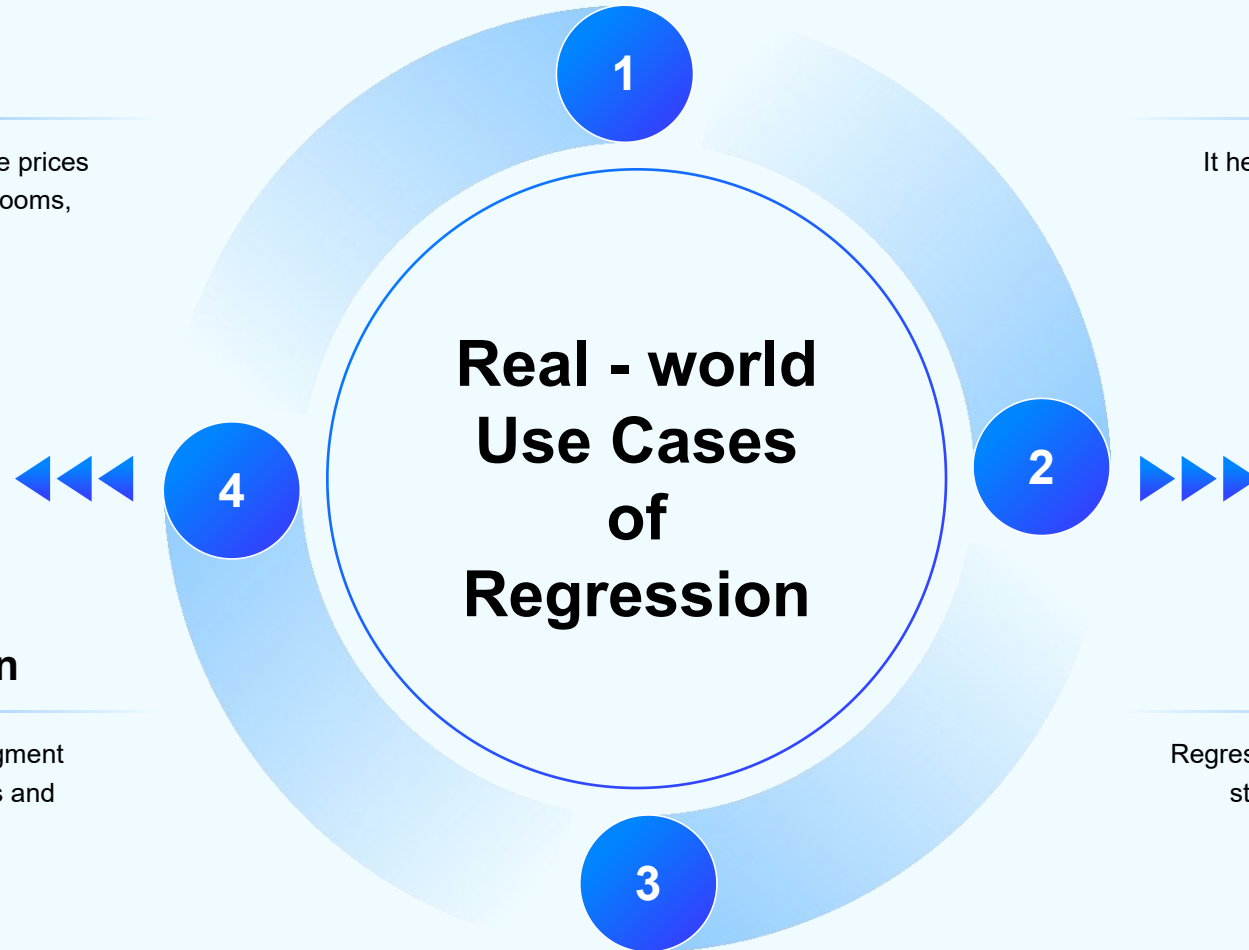
It helps businesses forecast sales by analyzing historical sales data and market trends.

## Customer Segmentation

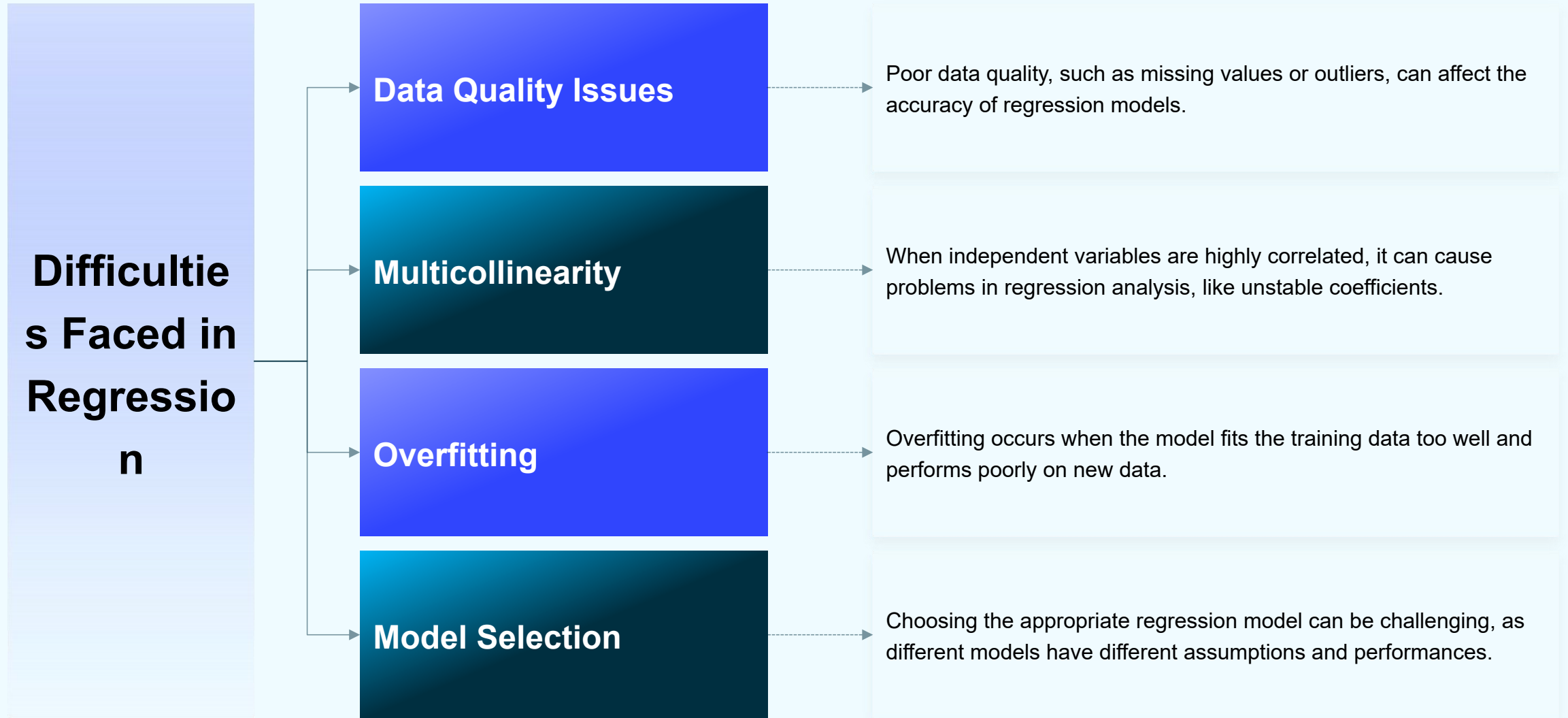
By using regression, companies can segment customers based on their characteristics and behavior.

## Stock Market Analysis

Regression is used in the stock market to analyze stock prices and make investment decisions.



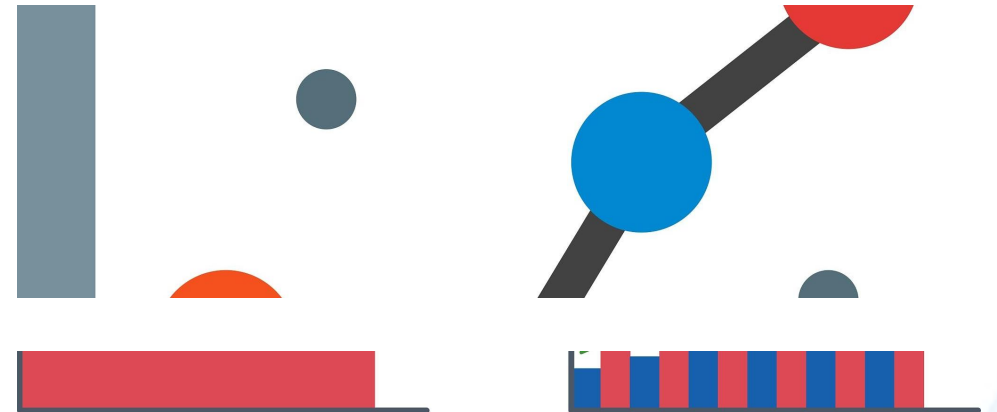
# Challenges in Regression



# Types of Regression Models

## Linear regression

Assuming a linear relationship between the target variable and the feature, optimize the parameters using the least squares method. Suitable for scenarios where the relationship between features and output is simple and satisfies linear assumptions, such as the relationship between advertising investment and sales revenue.



# Types of Regression Models

## Polynomial regression

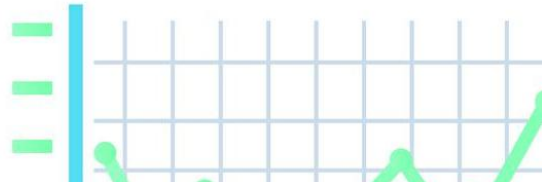
Fitting nonlinear relationships by introducing high-order terms of features (such as squares and cubes), but being cautious of overfitting. For example, parabolic data can be modeled using quadratic polynomials.

## Regularized Regression (Lasso/Ridge)

Adding L1 (Lasso) or L2 (Ridge) regularization terms to the loss function to prevent overfitting and improve generalization ability. Lasso can also achieve feature selection by compressing the coefficients of unimportant features to zero.



# Simple Linear Regression



## Basic Concept

Simple linear regression is a statistical method used to analyze the relationship between two continuous variables, where one variable (the dependent variable) is influenced by the other (the independent variable). The model is expressed as:

$$y = \beta_0 + \beta_1 x + \epsilon$$

where:

- $\beta_0$  = intercept,
- $\beta_1$  = slope,
- $\epsilon$  = error term.



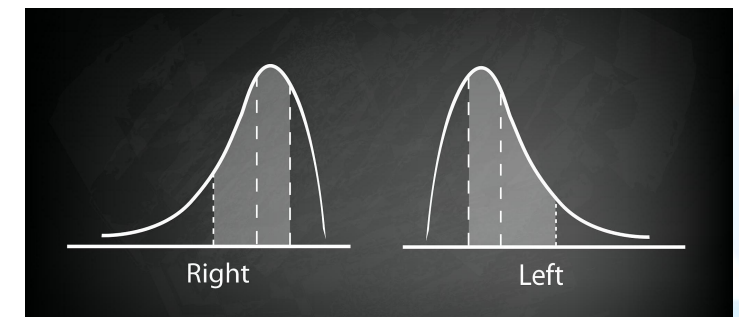
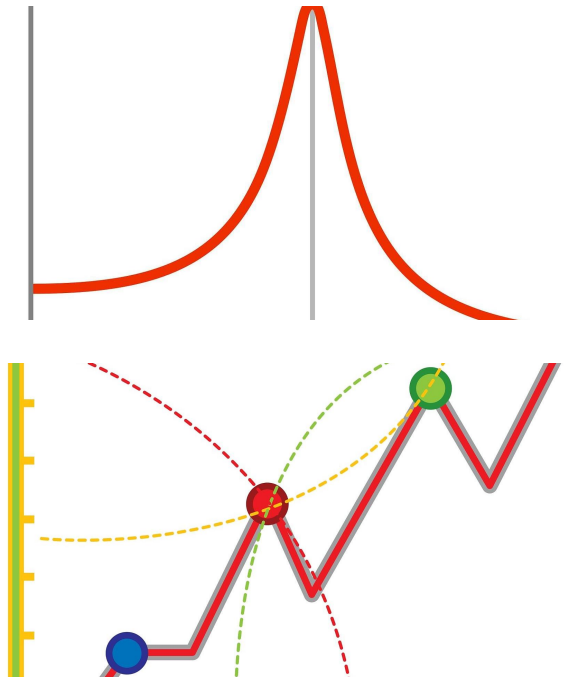
# Simple Linear Regression

## Application scenario

Suitable for exploring the influence of a single independent variable on the dependent variable, such as studying the relationship between advertising investment and sales, or the impact of temperature on ice cream sales.

## Advantages and disadvantages

The advantage is that the model is simple and easy to explain; The disadvantage is limited modeling ability for nonlinear or multivariate relationships, and sensitivity to outliers.



# Simple Linear Regression

## ⚙️ Core Assumption

$$y = X\beta + \varepsilon$$

where  $\varepsilon \sim N(0, \sigma^2)$

## Objective Function

$$\text{Minimize SSE} = \sum (y_i - \hat{y}_i)^2$$

## ★ Best-Case Scenario

- ✓ Low-dimensional data ( $p < n$ )
- ✓ Features linearly correlated with target

## 📋 Quick Checklist

- ✓ Homoscedastic residuals
- ✓ No multicollinearity
- ✓ Normally distributed errors

# Multiple Linear Regression

**Basic Concept:** Multiple linear regression extends simple linear regression by allowing the use of multiple independent variables to predict a single dependent variable. The model is represented by the equation:  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$ .

**Variable Selection:** Key variables must be identified using techniques like stepwise regression, LASSO, or Ridge regression. This prevents overfitting and reduces interference from irrelevant features.

**Interaction and Polynomial Terms:** To model nonlinear relationships, interaction terms (e.g.,  $X_1 \times X_2$ ) or polynomial terms (e.g.,  $X_1^2$ ) can be added to enhance the model's predictive flexibility.

**Application Cases:** Widely used in complex real-world scenarios such as:

- Housing price prediction (using area, location, age, etc.)
- Medical data analysis (e.g., assessing multi-factor disease risk)

# Assumptions and Limitations

## Linear assumption

It is required that there is a linear relationship between the dependent variable and the independent variable, otherwise it needs to be transformed (such as logarithmic transformation) or replaced with a nonlinear model.

## Independence of error terms

Error terms should be independent of each other and follow a normal distribution, otherwise it may lead to parameter estimation bias (such as time series data requiring special processing).

## Multicollinearity problem

High correlation between independent variables can reduce model stability, which can be detected and solved by variance inflation factor (VIF).

## Limitations

Sensitive to outliers and unable to directly handle categorical variables (requiring coding); Complex relationships may be replaced by models such as decision trees or neural networks.

# Polynomial Regression

## Model Form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d$$

## Power vs. Peril

$d = 2$ : Gentle curvature

$d \geq 5$ : Wild oscillations



## Overfitting Red Flags

↑ Training  $R^2$  ↑

↓ Validation  $R^2$  ↓

☒ Coefficients explode

## Defense Tactics

✓ Cross-validation

✓ Regularization



# Polynomial Regression

## 1. Nonlinear Data Fitting:

Extends linear regression by incorporating higher-order terms (e.g., quadratic, cubic). This effectively models nonlinear relationships and captures curved trends between features and target variables.

## 2. Feature Engineering:

Requires careful selection of polynomial order:

- **Underfitting:** Occurs with too-low orders.
- **Overfitting:** Risks from excessively high orders.

Optimal order is determined via cross-validation or validation-set error analysis.

## 3. Computational Complexity:

Feature dimensions expand (e.g.,  $x \rightarrow x^2, x^3$ ), increasing computational cost versus linear regression. High-dimensional data may necessitate optimized solution methods.

## 4. Regularization:

L1/L2 regularization (e.g., Ridge/Lasso) mitigates overfitting. In Scikit-Learn, implement via:

`PolynomialFeatures` → `Ridge` / `Lasso` pipeline.

# Logistic Regression Basics

- **Classification task core:** Although logistic regression is called "regression", it is actually a linear model for handling binary classification problems. The linear output is mapped to the  $[0,1]$  probability interval through the Sigmoid function.
- **Loss function characteristics:** Log loss is used as the optimization objective to penalize the deviation between predicted probabilities and true labels. Gradient descent or Newton's method are commonly used for parameter solving.
- **Multi class extension:** It can be extended to multi class tasks through One vs Rest (OvR) or Softmax regression, such as the multi class parameter configuration in Logistic Regression in Scikit Learn.
- **Strong interpretability:** The model coefficients directly reflect the impact of features on log Odds, which facilitates the analysis of feature importance in business scenarios, such as medical risk factor assessment.



# Summary of Regression Formulas

## 2.1 Simple Linear Regression

$$y = \beta_0 + \beta_1 x$$

- $\beta_0$ : intercept
- $\beta_1$ : slope

## 2.2 Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

- Multivariate feature combination

## 2.3 Polynomial Regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_k x^k$$

- Captures nonlinearity

## 2.3 Polynomial Regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_k x^k$$

- Captures nonlinearity

## 2.4 Logistic Regression (Probability)

$$\pi(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

- Binary classification

## 2.5 Cost Functions

- MSE: Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum (y - \hat{y})^2$$

- Linear regression

# Ridge and Lasso Regression



01

## Addressing multicollinearity

Ridge regression (L2 regularization) reduces the sensitivity of the model to highly correlated features by penalizing the sum of squared coefficients, making it suitable for scenarios where there is strong correlation between features.

02

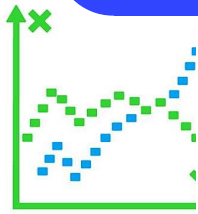
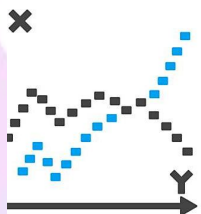
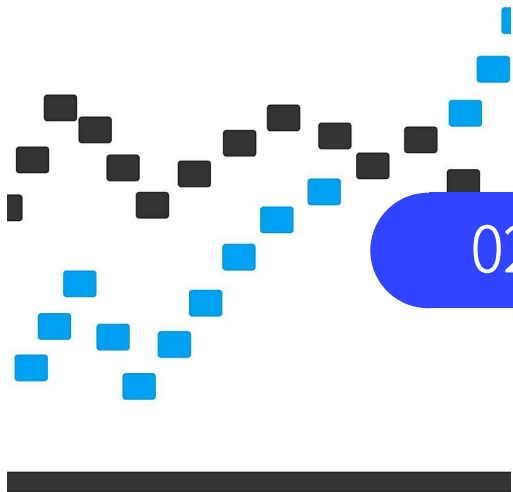
## Feature selection capability

Lasso regression (L1 regularization) can compress the coefficients of unimportant features to zero, achieving automatic feature selection, especially suitable for high-dimensional sparse data.

03

## Hyperparameter tuning

The regularization strength ( $\lambda$  or  $\alpha$ ) of both needs to be determined through grid search or cross validation, balancing bias and variance, and optimized through RidgeCV or LassoCV in Scikit Learn.



# Ridge Regression: Taming Multicollinearity with L2

## $\lambda$ Effect

$\lambda = 0$  → OLS

$\lambda \rightarrow \infty$  → All  $\beta_i \rightarrow 0$

## Geometric Insight



Circular constraint in parameter space

 Shrinks but never annihilates coefficients

## Effect of $\lambda$

- $\lambda = 0$ : No regularization → OLS.
- $\lambda \rightarrow \infty$ : All coefficients ( $\beta$ ) are pushed toward 0.
  - In Ridge Regression: Coefficients **shrink** but **never become exactly zero**.
  - In Lasso Regression: Some coefficients can be **exactly zero** → useful for **feature selection**.

## Circular Constraint in Parameter Space

- This refers to the **constraint region** imposed by regularization.
  - Ridge Regression uses an **L2 penalty**:  $\lambda \sum \beta_i^2 \rightarrow$  **Circular** constraint.
  - Lasso uses an **L1 penalty**:  $\lambda \sum |\beta_i| \rightarrow$  **Diamond-shaped** constraint.

# Ridge Regression

Comparing Ridge Regression to OLS and possibly Lasso.

Ridge adds a penalty to large coefficients, helping with multicollinearity and overfitting, but it doesn't eliminate features — it just shrinks them.

## Loss Function

The loss function is given as:

$$J(\Theta) = \text{MSE} + \lambda \cdot \text{Regularization Term}$$

- **MSE:** Mean Squared Error — measures how well the model predicts the data.
- **$\lambda$  (lambda):** A regularization parameter that controls the strength of the penalty on model complexity.

## Geometric Insight

This likely refers to how regularization affects the **parameter space**:

- **Without regularization ( $\lambda = 0$ ):** This is **Ordinary Least Squares (OLS)** — no penalty, so coefficients are free to grow large.
- **With regularization ( $\lambda > 0$ ):**
  - Coefficients are **shrunk** toward zero.
  - The constraint region (e.g., circle for Ridge, diamond for Lasso) restricts the values of  $\beta$  (coefficients).
  - This helps reduce **overfitting** and improves **generalization**.

# Lasso Regression: Feature Selection via L1 Magic

## Loss Function

$$J(\theta) = \text{MSE} + \lambda \sum |\beta_i|$$

## $\lambda$ Effect

Forces exact zeros  $\rightarrow$  automatic feature pruning

Active Features:



30% remaining

## Coordinate Descent Snapshot

| Iteration | Active Features | Validation RMSE |
|-----------|-----------------|-----------------|
| 0         | 50              | 3.21            |
| 20        | 12              | 2.87            |
| 100       | 7               | 2.85            |



70% convergence

## Ideal Use Case

  $p \gg n$

Genomics, text data

 Interpretability

Mission-critical

# Implementation : Data Preprocessing for Regression

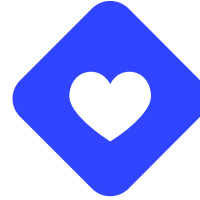
- **Missing value handling:** Fill in missing data with mean, median, or mode, or use interpolation, deletion, and other methods to ensure data integrity. For time series data, forward or backward filling strategies can be used.
- **Data standardization/normalization:** Scaling numerical features to the same dimension (such as Z-score normalization or Min Max normalization) to avoid model bias caused by significant numerical differences and improve gradient descent efficiency.
- **Outlier detection and handling:** Identify outliers using box plots, IQR, or  $3\sigma$  principles, and optimize data distribution through truncation, substitution, or preservation (such as tree models that may not require processing).
- **Category feature encoding:** Use one hot encoding, label encoding, or target encoding for categorical variables, and convert them into numerical form for model processing.

# Implementation : Feature Selection Methods



## Filter method

Based on statistical indicators such as variance, chi square test, and mutual information, filter features with high computational efficiency but ignoring model characteristics. For example, removing low variance features with variance close to zero.



## Wrapper method

By using recursive feature elimination (RFE) or forward/backward selection, combined with model performance iteration to select the optimal feature subset, the computational cost is high but the effect is significant.



## Embedded method

Utilizing the built-in feature importance of the model (such as coefficient shrinkage in Lasso regression and Gini importance in random forests) to automatically select features, balancing efficiency and accuracy.

# Implementation : Hyperparameter Tuning

- **Grid Search:** Traverse pre-defined hyperparameter combinations and evaluate performance through cross validation, suitable for small-scale parameter spaces but with high computational resource consumption.
- **Random Search:** Randomly sampling combinations within a parameter range, more efficient than grid search, especially suitable for high-dimensional parameter spaces.
- **Bayesian optimization:** Based on Gaussian process or TPE algorithm to model the objective function, intelligently explore the hyperparameter space, and find the optimal solution with fewer iterations, suitable for expensive model optimization.
- **Early Stopping mechanism:** monitors the loss of the validation set during iterative training, terminates training when performance no longer improves, prevents overfitting and saves time. It is commonly used in gradient boosting trees or neural networks.



# Case Studies : Predicting Housing Prices

1

## Core business value

Real estate price prediction is a key tool in the fields of finance, investment, and urban planning. Accurate models can help investors optimize asset allocation and government agencies formulate reasonable housing policies.

2

## Feature engineering challenge

Dealing with multiple features (such as geographic location, number of rooms, school district ratings, etc.), involving nonlinear relationship modeling and missing data filling, testing data cleaning and feature selection abilities.

3

## Model generalization requirements

The market dynamics in different regions vary significantly, requiring the model to have the ability to adapt to local trends, such as avoiding overfitting through regularization or ensemble learning.



# Boston Housing Price Prediction Steps

## Predicting Boston House Prices with Gradient Boosting

### Imports

Import necessary libraries like numpy, pandas, scikit - learn, etc.

### Load Data

Load the Boston housing dataset from the openml source.

### Split Data

Split the data into train, validation, and test sets using appropriate methods.

### Create Pipeline

Build a pipeline with scaling and Gradient Boosting Regressor.

### Define Hyper - parameter Grid

Set up a grid of hyper - parameters for the GridSearchCV.

### Fit and Evaluate

Fit the model using GridSearchCV and evaluate on the test set.

### Check Residuals

Plot the residuals vs. fitted values to analyze the model's performance.

# Case Studies : Stock Market Forecasting



## Data complexity

It is necessary to integrate multiple sources of macroeconomic indicators (such as GDP, interest rates), news sentiment analysis, etc., and handle high-frequency noise and non-stationary time series.

## Risk control

The model needs to quantify prediction uncertainty (such as Bayesian regression), dynamically adjust positions in combination with stop loss strategies, and avoid systemic risks caused by black swan events.



## Real time requirements

Deploy lightweight models (such as linear regression variants) to meet low latency trading needs, while regularly updating parameters online to adapt to market changes.



# Case Studies : Medical Diagnosis Using Regression

- By regression analysis of the relationship between patient physiological indicators (blood pressure, cholesterol, etc.) and disease incidence, an early warning model is constructed to assist clinical triage decision-making.
- To address the issue of data imbalance (such as a small sample size in rare cases), weighted loss functions or oversampling techniques should be used to improve model sensitivity.
- Establish a dose-response regression model to quantify the relationship between drug dosage and efficacy/side effects, and support the development of personalized treatment plans.
- When dealing with confounding variables such as patient age and complications, it is necessary to introduce stratified regression or causal inference methods to ensure the reliability of the conclusions.

# Comparison of Basic Regressions

| Criterion                  |  Linear |  Polynomial |  Ridge |  Lasso |
|----------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Linearity Assumption       | ✓                                                                                        | ✗                                                                                              | ✓                                                                                         | ✓                                                                                         |
| Built-in Feature Selection | ✗                                                                                        | ✗                                                                                              | ✗                                                                                         | ✓                                                                                         |
| Handles Multicollinearity  | ✗                                                                                        | ✗                                                                                              | ✓                                                                                         | ✗                                                                                         |
| Risk of Overfitting        | Low                                                                                      | High                                                                                           | Medium                                                                                    | Medium                                                                                    |
| Interpretability           | High                                                                                     | Medium                                                                                         | High                                                                                      | High                                                                                      |

# Common Pitfalls in Regression

## Identifying and Mitigating Issues

### 1 Data Leakage

Scaling on the whole dataset before splitting can inflate the score.

### 2 Extrapolation

Tree - based models predict flatly beyond the training data range.

### 3 Zero - inflated Targets

Consider log - transform or Tweedie loss for such targets.

### 4 High Cardinality Categoricals

Use target encoding or CatBoost to handle high - cardinality categoricals.

### 5 Ignoring Interaction Terms

Don't overlook interaction terms suggested by physics or domain knowledge.

### 6 Using ( $R^2$ ) Alone

(  $R^2$  ) can be high even when absolute errors are unacceptable.

# Deeper Dives in Regression

## Advanced Concepts and Tools

- **Regularized GLMs**

Use Poisson, Gamma, or Tweedie regression for specific data types.

- **Quantile Regression**

Implement quantile regression for prediction intervals using Gradient Boosting.

- **Bayesian Regression**

Explore Bayesian regression with tools like PyMC, Stan, or TensorFlow Probability.

- **AutoML**

Utilize AutoML tools like auto - sklearn, H2O, or AutoGluon for baseline models.

- **Explainability**

Use SHAP values for tree models and coefficients for linear models.

- **Deployment**

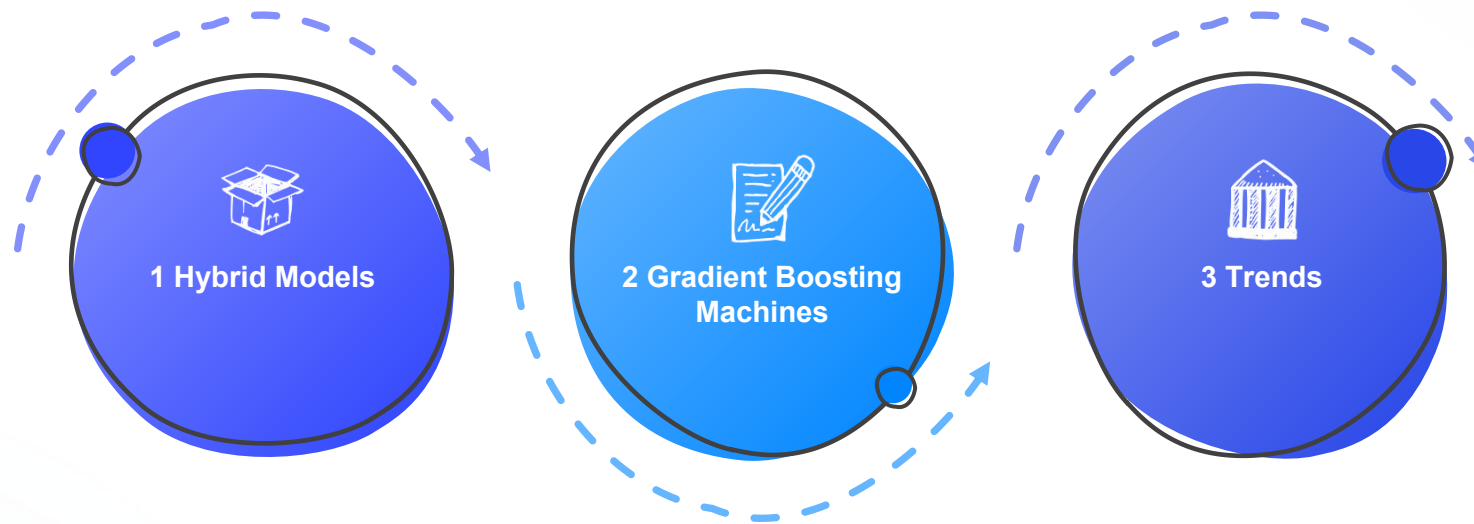
Deploy models as REST APIs with FastAPI or perform batch inference with Spark.

# Comparison of Regression Algorithms

| Algorithm                                   | Description                                   | Use Case                 |
|---------------------------------------------|-----------------------------------------------|--------------------------|
| Linear Regression                           | Fits a straight line to the data              | House price prediction   |
| Polynomial Regression                       | Fits a curve using polynomial features        | Temperature trends       |
| Ridge / Lasso Regression                    | Adds regularization to prevent overfitting    | High-dimensional data    |
| Decision Tree Regression                    | Uses tree splits to predict continuous values | Non-linear relationships |
| Random Forest Regression                    | Ensemble of decision trees                    | Robust predictions       |
| Support Vector Regression (SVR)             | Uses SVM principles for regression            | Small datasets           |
| Gradient Boosting (e.g., XGBoost, LightGBM) | Boosts weak learners sequentially             | Kaggle competitions      |



# Advanced Topics & Trends



Ridge Regression + Neural Networks  
Case Study: Carcinoma identification (97.69% accuracy)

XGBoost, LightGBM: State-of-the-art for tabular data

Integration of traditional algorithms with deep learning  
AutoML for regression hyperparameter tuning



# Regression vs Gradient Descent

A Comprehensive Comparison

# Gradient Descent

## What is Gradient Descent?

1

### Definition

Gradient descent is an optimization algorithm mainly used to minimize a function. It adjusts parameters step - by - step to gradually reduce the function value until reaching the minimum.

2

### Basic Idea

The core idea is to calculate the gradient of the function, which is the partial derivative. Then, it adjusts parameters according to the gradient direction to approach the minimum.

3

### Application Scenarios

It is widely used in fields like machine learning, data science, and artificial intelligence, playing a key role in training models and optimizing algorithms.

4

### Historical Development

Although the exact origin time is uncertain, gradient descent has been continuously improved and widely applied in mathematical and computer - science fields over the years.

# Regression Analysis

## An Overview of Regression

### Definition

Regression analysis is a statistical method for prediction and modeling. It builds a model to describe the relationship between one or more dependent variables and some independent variables.

### Main Goals

The goal is to find the best model that makes the relationship between the dependent and independent variables the most significant.

### Types of Regression

There are various types of regression, including linear regression, quadratic regression, locally weighted regression, logistic regression, etc., each suitable for different data scenarios.

### Role in Data - driven Decisions

Regression analysis helps in making data - driven decisions by predicting future values and understanding the underlying relationships in data, such as market trend prediction and financial risk assessment.

# Gradient Descent Concepts

## Function

Gradient descent aims to minimize a function. The choice and definition of the function are crucial for the algorithm's performance, as different functions have different optimization difficulties.

## Learning Rate

The learning rate is a parameter in gradient descent that controls the speed of gradient descent. A too - large learning rate may cause the parameters to oscillate around the minimum, while a too - small one may slow down the optimization process.

## Convergence

Gradient descent is designed to converge to the minimum of the function under certain conditions. However, the convergence speed and stability can be affected by factors like the learning rate and the shape of the function.

## Components of Gradient Descent

### Gradient

The gradient is the partial derivative of the function, indicating the rate of growth of the function at a certain point. It serves as the key guidance for parameter adjustment in gradient descent.

### Cost Function

The cost function is used to measure the difference between the predicted values of the model and the actual values. Gradient descent aims to minimize the cost function to optimize the model parameters.

# Gradient Descent Formula

**Context:** Linear regression with model  $h_{\theta}(x) = \theta_0 + \theta_1 x$

**Goal:** Minimize Mean Squared Error (MSE) loss:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

**Parameter Update Rule:**

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

- $\theta_j$ : Parameter (weight/bias)
- $\alpha$ : Learning rate
- $\frac{\partial J}{\partial \theta_j}$ : Gradient of loss w.r.t.  $\theta_j$

**Gradients for Linear Regression:**

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

# Regression Analysis Concepts

## Elements of Regression

### Dependent Variable

The dependent variable is the target variable in regression analysis that needs to be predicted or modeled. Its selection depends on the research purpose, such as predicting housing prices or stock returns.

### Independent Variable

The independent variable is the input variable in regression analysis, and the relationship between it and the dependent variable needs to be modeled. The choice of independent variables can significantly impact the model's accuracy and reliability.

### Model

Regression analysis builds a model to describe the relationship between the dependent and independent variables. Different models have different assumptions and 适用场景s, like linear models for simple relationships and nonlinear models for complex ones.

### Residuals

Residuals are the differences between the observed values of the dependent variable and the predicted values by the regression model. They are used to evaluate the goodness - of - fit of the model and detect potential errors or outliers.

### Correlation

Correlation measures the degree of association between the dependent and independent variables in regression analysis. A high correlation indicates a strong relationship, which can help in building more accurate prediction models.

# Similarities and Differences

## Comparing the Two

### Conceptual Similarities

Both gradient descent and regression analysis involve function minimization and modeling to some extent. Gradient descent minimizes a cost function, while regression analysis builds a model to minimize prediction errors.

### Goal Differences

The goal of gradient descent is to find the minimum value of a function, while the goal of regression analysis is to find the best model to describe the relationship between variables.

### Application Differences

Gradient descent is mainly used in optimization algorithms, while regression analysis is used for statistical modeling and prediction. They have different application fields, such as algorithm optimization and data - driven decision - making.

### Suitability in Different Scenarios

Gradient descent is more suitable for parameter optimization problems, while regression analysis is better for data - driven prediction and modeling tasks. The choice depends on the specific problem at hand.





# Key Difference Summary

| Feature  | Regression                            | Gradient Descent                       |
|----------|---------------------------------------|----------------------------------------|
| Type     | Model / Technique                     | Optimization Algorithm                 |
| Purpose  | Predict or model relationships        | Minimize a function                    |
| Output   | Model parameters (e.g., coefficients) | Optimized parameter values             |
| Used for | Prediction, inference                 | Training models (including regression) |
| Example  | Linear regression model               | Used to fit linear regression          |

# Gradient Descent Algorithm

## How Gradient Descent Works



Core Principle

1

Operation Steps

2

Mathematical Formulas

3

Learning Rate Adjustment Strategies

4

Algorithm Advantages and Disadvantages

5

# Regression Analysis Model

## Modeling in Regression

1

### Mathematical Models

Regression analysis has various mathematical models, such as linear regression models like  $y = h(x) = w * x + b$ , and quadratic regression models like  $y = h(x) = a * x^2 + b * x + c$ .

2

### Housing Price Estimation Example

For example, in a housing price estimation system, assume the house value is only affected by its area. We can use a linear regression model to fit the data and predict the price of a new house based on its area.

4

### Model Limitations

Regression models have limitations, such as their ability to handle nonlinear relationships and sensitivity to outliers. These limitations need to be considered when applying regression analysis in practice.

3

### Model Evaluation Metrics

Common evaluation metrics for regression models include mean - squared error (MSE), mean absolute error (MAE), and R - squared ( $R^2$ ). These metrics help in assessing the model's accuracy and performance.

# Summary

## Key Takeaways

### 1 Key Points of Gradient Descent

Gradient descent is an optimization algorithm that minimizes functions through parameter adjustment. It has important applications in machine learning and other fields.

### 2 Key Points of Regression Analysis

Regression analysis is a statistical method for prediction and modeling, aiming to find the best model to describe variable relationships.

### 3 Similarities and Differences Recap

Both involve function minimization and modeling, but have different goals and applications. Gradient descent focuses on function optimization, while regression analysis focuses on building relationship models between variables.

# Comparison

## Comparing the Two Methods

### Similarities in Concept and Application

They both use mathematical models to solve problems and have some overlap in their concepts, such as function minimization. However, their specific applications and goals are different.

### Differences in Goals and Scenarios

Gradient descent aims at function minimization, while regression analysis aims at building predictive models. They are suitable for different problem scenarios, like algorithm optimization and data - driven prediction.

### Complementary Nature

In some cases, gradient descent can be used to optimize the parameters of a regression model, showing a complementary relationship between the two methods.

# Outlook

## Future Directions

### 1 Future Development Trends of Gradient Descent

With the development of artificial intelligence and big data, gradient descent will continue to be improved, such as in the development of more efficient optimization algorithms.

### 2 Future Development Trends of Regression Analysis

Regression analysis will also evolve, with the emergence of new types of regression models and the integration of more advanced technologies.

### 3 Potential Applications in New Fields

Both gradient descent and regression analysis may find new applications in emerging fields like quantum computing and bioinformatics, driving the development of these fields.